detailed picture of what the neuronal hardware of a meme might look like. Among the other interesting things he does is to explore, far more searchingly than I had done, the analogy of memes with parasites; to be more precise, with the spectrum of which malignant parasites are one extreme, benign 'symbionts' the other extreme. I am particularly keen on this approach because of my own interest in 'extended phenotypic' effects of parasite genes on host behaviour (see Chapter 13 of this book and especially chapter 12 of *The Extended Phenotype*). Delius, by the way, emphasizes the clear separation between memes and their ('phenotypic') effects. And he reiterates the importance of coadapted meme complexes, in which memes are selected for their mutual compatibility.

p. 252 'Auld Lang Syne'

'Auld Lang Syne' was, unwittingly, a revealingly fortunate example for me to have chosen. This is because, almost universally, it is rendered with an error, a mutation. The refrain is, essentially always nowadays, sung as 'For the sake of auld lang syne', whereas Burns actually wrote 'For auld lang syne'. A memically minded Darwinian immediately wonders what has been the 'survival value' of the interpolated phrase, 'the sake of'. Remember that we are not looking for ways in which *people* might have survived better through singing the song in altered form. We are looking for ways in which the alteration *itself* might have been good at surviving in the meme pool. Everybody learns the song in childhood, not through reading Burns but through hearing it sung on New Year's Eve. Once upon a time, presumably, everybody sang the correct words. 'For the sake of' must have arisen as a rare mutation. Our question is: why has the initially rare mutation spread so insidiously that it has become the norm in the meme pool?

I don't think the answer is far to seek. The sibilant 's' is notoriously obtrusive. Church choirs are drilled to pronounce 's' sounds as lightly as possible, otherwise the whole church echoes with hissing. A murmuring priest at the altar of a great cathedral can sometimes be heard, from the back of the nave, only as a sporadic sussuration of 's's. The other consonant in 'sake', 'k', is almost as penetrating. Imagine that nineteen people are correctly singing 'For auld lang syne' and one person, somewhere in the room, slips in the erroneous 'For the sake of auld lang syne'. A child, hearing the song for the first time, is eager to join in but uncertain of the words. Although almost everybody is singing 'For auld lang syne', the hiss of an 's' and the cut of a 'k' force their way into the child's ears, and when the refrain comes round again he too sings 'For the sake of auld lang syne'. The mutant meme has taken over another vehicle. If there are any other children there, or adults unconfident of the words, they will be more likely to switch to the mutant form next time the refrain comes round. It is not that they 'prefer' the mutant form. They genuinely don't know the words and are honestly eager to learn them. Even if those who know better indignantly bellow 'For auld lang syne' at the top of their voice (as I do!), the correct words happen to have no conspicuous consonants, and the mutant form, even if quietly and diffidently sung, is far easier to hear.

A similar case is 'Rule Britannia'. The correct second line of the chorus is 'Britannia, rule the waves.' It is frequently, though not quite universally, sung as 'Britannia rules the waves.' Here the insistently hissing 's' of the meme is aided by an additional factor. The intended meaning of the poet (James Thompson) was presumably imperative (Britannia, go out and rule the waves!) or possibly subjunctive (let Britannia rule the waves). But it is superficially easier to misunderstand the sentence as indicative (Britannia, as a matter of fact, does rule the waves). This mutant meme, then, has two separate survival values over the original form that it replaced: it sounds more conspicuous and it is easier to understand.

The final test of a hypothesis should be experimental. It should be possible to inject the hissing meme, deliberately, into the meme pool at a very low frequency, and then watch it spread because of its own survival value. What if just a few of us were to start singing 'God saves our gracious Queen'?